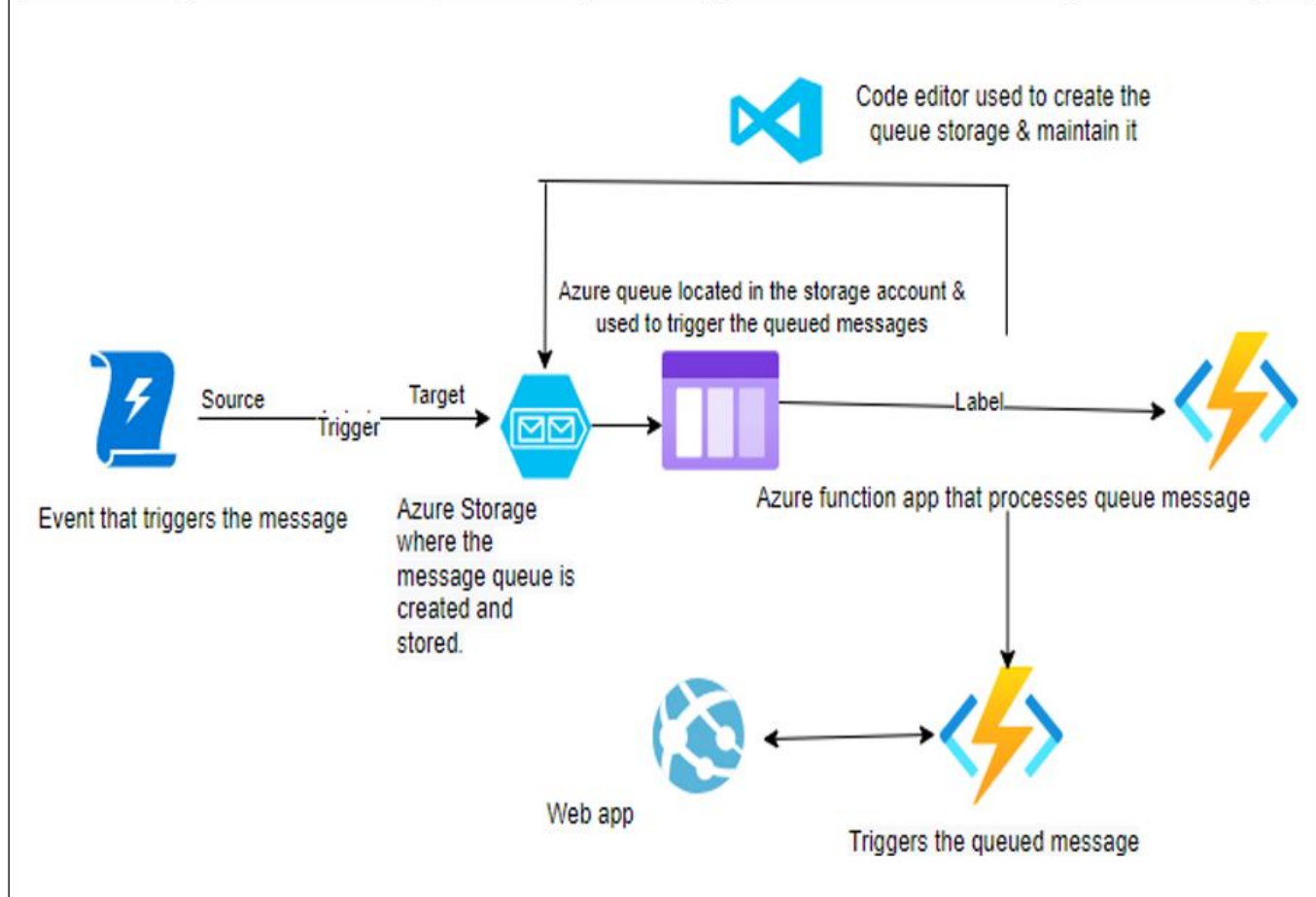


Use Case 2: Azure Capstone 7: Create a function triggered by Azure Queue storage

2.1 Solution Diagram—

Solution diagram to use Azure queue storage that triggers function call to send queued messages.



2.3 Steps Applied—

Step1: Creation of resource group. Resource group name- **GLCapstoneRG**

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

suchitra.singh2@gmail.com...
DEFAULT DIRECTORY (SUCHITRA...)

Home > Resource groups >

Resource groups

Default Directory (suchitrasingh2@gmail.com@microsoft...)

+ Create Manage view ...

Filter for any field...

Name ↑

GLCapstoneRG

GLCapstoneRG

Resource group

Search

+ Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete ...

JSON View

Essentials

Subscription (move) : [Azure subscription 1](#) Deployments : [No deployments](#)

Subscription ID : a644ae14-f645-4315-997d-0ab78887eb78 Location : East US 2

Tags (edit) : [Add tags](#)

Resources

Recommendations (1)

Filter for any field... Type equals all Location equals all Add filter

Showing 0 to 0 of 0 records. ☐ Show hidden types

No grouping List view

Name ↑	Type ↑	Location ↑
--------	--------	------------

No resources match your filters.
Try changing or clearing your filters.

Create resources Clear filters

Learn more Give feedback

Step 2: Creation of a storage account. Name- mystoragequeue1234

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

suchitra.singh2@gmail.com...
DEFAULT DIRECTORY (SUCHITRA...)

me > mystoragequeue1234_1737236753334 | Overview >

mystoragequeue1234

Storage account

Search

Upload Open in Explorer Delete Move Refresh Open in mobile CLI / PS Feedback

JSON View

Essentials

Resource group (move)	: GLCapstoneRG	Performance	: Standard
Location	: eastus2	Replication	: Locally-redundant storage (LRS)
Subscription (move)	: Azure subscription 1	Account kind	: StorageV2 (general purpose v2)
Subscription ID	: a644ae14-f645-4315-997d-0ab78887eb78	Provisioning state	: Succeeded
Disk state	: Available	Created	: 1/18/2025, 4:46:00 PM
Tags (edit)	: Add tags		

Properties

Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

Blob service

Hierarchical namespace	Disabled
Default access tier	Hot
Blob anonymous access	Disabled
Blob soft delete	Enabled (7 days)
Container soft delete	Enabled (7 days)
Versioning	Disabled
Change feed	Disabled
NFS v3	Disabled
Allow cross-tenant replication	Disabled
Storage tasks assignments	None

Security

Require secure transfer for REST API operations	Enabled
Storage account key access	Enabled
Minimum TLS version	Version 1.2
Infrastructure encryption	Disabled

Networking

Allow access from	All networks
Private endpoint connections	0
Network routing	Microsoft network routing
Access for trusted Microsoft services	Yes
Endpoint type	Standard

File service

Step 3: Adding a queue to the storage account. Queue Name-**myqueue**

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

suchitra.singh2@gmail.c...
DEFAULT DIRECTORY (SUCHITRA...)

Home > mystoragequeue1234_1737236753334 | Overview > mystoragequeue1234

mystoragequeue1234 | Queues ✕ ☆ ...

Storage account

Search

+ Queue Refresh Delete Give feedback

Search queues by prefix

Queue	Url
myqueue	https://mystoragequeue1234.queue.core.windows.net/myqueue

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Step 4: Creation of a function app. Function app name-**capstonefunctionapp**

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

suchitra.singh2@gmail.c...
DEFAULT DIRECTORY (SUCHITRA...)

Home > Microsoft.Web-FunctionApp-Portal-3e64730d-acf2 | Overview >

capstonefunctionapp ✕ ☆ ...

Function App

Search

Browse Refresh Stop Restart Swap Get publish profile Reset publish profile Download app content Delete Send us your feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

Recommended services (preview)

Functions

Deployment

Settings

Performance

App Service plan

Development Tools

API

Monitoring

Automation

Support + troubleshooting

Essentials

Resource group (move) : GICapstoneRG

Status : Running

Location (move) : East US 2

Subscription (move) : Azure subscription 1

Subscription ID : a644ae14-f645-4315-997d-0ab78887eb78

Tags (edit) : Add tags

Default domain : capstonefunctionapp.azurewebsites.net

Operating System : Windows

App Service Plan : ASP-GICapstoneRG-92dd (v1:0)

Runtime version : 4.1036.3.23284

JSON View

Functions Metrics Properties Notifications (0)

Create functions in your preferred environment

Create with Visual Studio

Best optimized for:

- Local development within Visual Studio
- Using a full .NET IDE

Create with Visual Studio

VS Code Desktop

Best optimized for:

- Local development within VS Code
- Custom development tool requirements

Create with VS Code Desktop

Other editors or CLI

Best optimized for:

- Using preferred editor for development
- Visual Studio, IntelliJ, command line

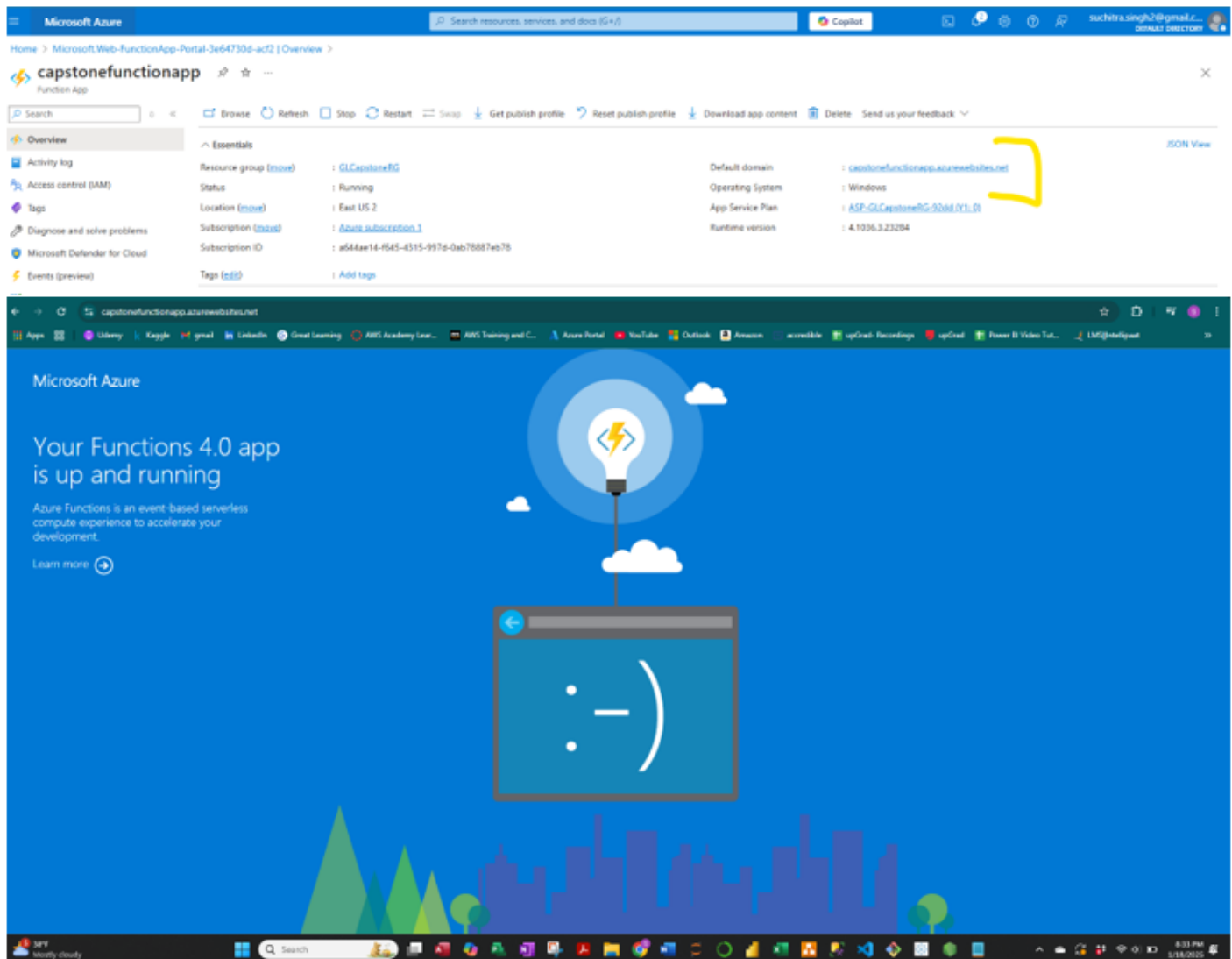
Set up your editor

38°F Mostly cloudy

Search

8:32 PM 1/18/2023

The following screenshot confirms that the app is running



Name of the function given in VS Code Studio- **HelloFunction**, namespace defined= **queuetrigger.Function**

Step 5: Creating the queue trigger in VS Code Studio

The screenshot shows the VS Code Studio interface with the following components:

- EXPLORER:** Shows the file structure of the project, including `function_app.py` and `Preview queue_trigger_template.md`.
- EDITOR:** Displays the `function_app.py` file with the following code:


```
1 import azure.functions as func
2 import logging
3
4 app = func.FunctionApp()
5
6 @app.queue_trigger(arg_name="azqueue", queue_name="myqueue",
7                   connection="mystoragequeue1234_STORAGE")
8 def QueueTriggerFunction(azqueue: func.QueueMessage):
9     logging.info('Python Queue trigger processed a message: %s',
10                 azqueue.get_body().decode('utf-8'))
```
- Preview queue_trigger_template.md:** Shows the template for the queue trigger, including the `QueueTriggerFunction` definition and the `QueueTriggerFunction` decorator.
- TERMINAL:** Shows the output of the `func host start` command, indicating that the function app is running successfully.

Step 6: Running the queue trigger in VS Code Studio

The screenshot shows the VS Code Studio interface with the following components:

- EDITOR:** Displays the `launch.json` file with the following configuration:


```
1 {
2   "version": "0.2.0",
3   "configurations": [
4     {
5       "name": "Attach to Python Functions",
6       "type": "debugpy",
7       "request": "attach",
8       "connect": {
9         "host": "localhost",
10        "port": 9091
11      },
12      "preLaunchTask": "func: host start"
13    }
14  ]
15 }
```
- TERMINAL:** Shows the output of the `func host start` command, indicating that the function app is running successfully. A yellow checkmark is placed next to the `QueueTriggerFunction: queueTrigger` line.
- CALL STACK:** Shows the stack trace of the `QueueTriggerFunction` function.
- BREAKPOINTS:** Shows the breakpoints for the `QueueTriggerFunction` function.
- RIGHT SIDE:** Shows the `azureFunctions.deploySubPath` settings, indicating that the default subpath of a workspace folder is used when deploying.

Step 7: Deploying the function app to Azure

The screenshot displays the Visual Studio Code interface during the deployment of a function app to Azure. The file explorer on the left shows the project structure, including the 'QueueTriggerFunction' folder. The code editor in the center shows the 'launch.json' file with the following configuration:

```
1 {
2   "version": "0.2.0",
3   "configurations": [
4     {
5       "name": "Attach to Python Functions",
6       "type": "debugpy",
7       "request": "attach",
8       "connect": {
9         "host": "localhost",
10        "port": 9091
11      },
12      "preLaunchTask": "func: host start"
13    }
14  ]
15 }
```

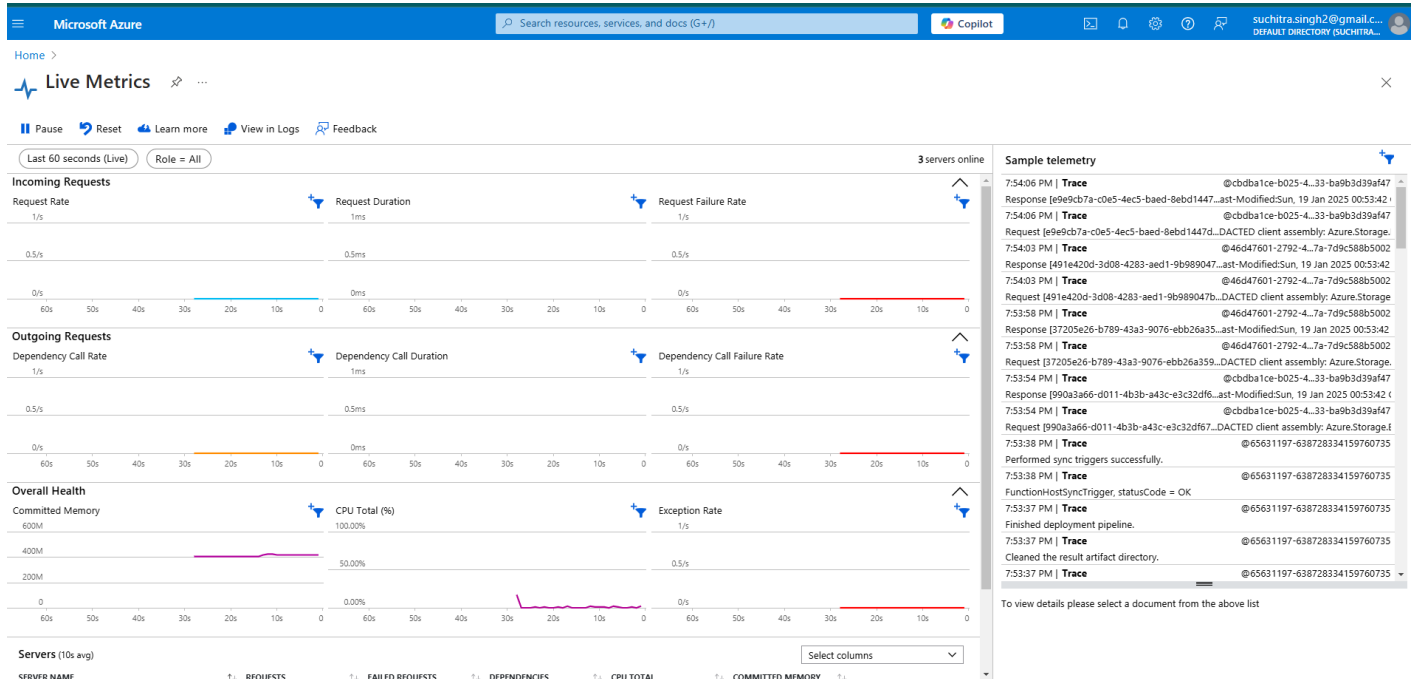
A dialog box prompts the user: "Are you sure you want to deploy to 'capstonefunctionapp'? This will overwrite any previous deployment and cannot be undone." The "Deploy" button is highlighted.

The terminal output shows the following messages:

```
ms-dotnettools.csharp.
Could not find the architecture of the dotnet host C:\Program Files\dotnet\dotnet.exe. If this host does not match the architecture x64:
Please set the PATH to a dotnet host that matches the architecture x64. An incorrect architecture will cause instability for the extension
ms-dotnettools.csharp.
ms-dotnettools.csharp requested to download the .NET Runtime.
Downloading .NET version(s) 8.0.12~x64 ...ms-dotnettools.csharp tried to install .NET 8.0.12~x64 but that install had already been
requested. No downloads or changes were made.
..... Done!
.NET 8.0.12~x64 executable path: c:\Users\suchi\AppData\Roaming\Code\User\globalStorage\ms-dotnettools.vscode-dotnet-runtime\.dotnet\8.0.
12~x64\dotnet.exe
```

The bottom status bar indicates "Deploy to App 'capstonefunctionapp' Succeeded". A yellow checkmark is drawn over the success message. The bottom right corner shows a notification: "Deployment to 'capstonefunctionapp' completed. Source: Azure Functions. Stream logs. Upload settings. View output."

Logstream confirming real-time logging of the app



2.4 Cost Analysis—

The cost estimate is based on usage of Azure resources for US East 2 region. Pricing information please visit <https://azure.microsoft.com/pricing/calculator/>

Azure Services	Cost Estimate
Azure Functions Memory size=128 Execution time (in milliseconds)=100 Requests per month = 100	\$0.00
Storage Accounts Block Blob Storage Redundancy=LRS Capacity=100 GB Plan Option=Pay-as-you-go	\$1.95 average per month
Operations and Data Transfer=10*10,000 operations List and Create Container Operations=10*10,000 operations Read operations=10*10,000 operations All other operations=1*10,000 operations Data Retrieval=100 GB= 100*\$0.00	\$0.050 (per 10,000 operations) =\$0.50 \$0.050 (per 10,000 operations) =\$0.50 \$0.004 (per 10,000 operations) =\$0.04 \$0.004 (per 10,000 operations) =\$0.01 \$0.00 Upfront Cost=\$0.00 Monthly Cost = \$2.88
App Service B1 Instance using 1.75GB RAM, 10GB storage@0.075 for 100 hours	\$7.50 monthly
Git Services (optional)	

Developer tools Support (<i>for atleast 2 developers</i>)	\$21.25 monthly \$29.00 monthly
Basic tools Support per user	\$0.00 monthly \$21.00
Total Estimate (Monthly) including Git Services = \$60.63 (<i>with Developer support</i>) \$ 31.63 (<i>with Basic support</i>)	
Total Estimate (Monthly) excluding Git Services = \$12.33	

2.5 Observations—

1. Region East US is more expensive than East US 2 specifically while using storage accounts.
2. East US 2 for a Flex Consumption hosting costs \$1.84 average per month while East US costs \$2.08 for a 100 GB capacity. Alternative application for a CI/CD pipeline other than Git services. The alternative to Git could be Azure DevOps.
3. It's advisable to select Locally Redundant Storage (LRS) over Geo-Redundant Storage (GRS) if the queue management is local.

-----End of Use Case 2-----
